Anthony (Youwei) Zhen

69 Brown St, Providence, Rhode Island
917-882-0066 | youweizhen.com | anthonyzhen0066@gmail.com | linkedin.com/in/youwei-zhen | github.com/AntoDono

Education

Brown University May 2028
Bachelor of Science in Applied Mathematics and Computer Science, ***GPA: 4.0***

Experience

Amazon NYC, New York
Software Development Engineering Intern *September 2026 - December 2026*
- Intern at AWS Athena and Glue ETL team.

SpaceX Starbase, Texas
Software Engineering Intern - Starship Design Software *June 2026 – September 2026*
- Built and shipped a **design-review system** to a **5,000+ person engineering organization**. In **1 month**, **165 unique engineers** logged **4,000+ feedback items** and **900+ comments**.
- Architected the **commenting stack**, including a **notification system**, autocompleted **@-mentions**, **file attachment previews**, and **editor integration**.
- Engineered an in-house **patch layer** over the **pdf.js engine**, rewriting it in **TypeScript** to add **spatial annotation** on **PDF drawings** and **3D models** to auto-generate screenshots pinpointing flagged components.
- Scaled **distributed job dispatch** to render **CAD drawings** on every design revision via concurrent worker queues.
- Collaborated with **10+ stakeholders** to drive a **10x company-wide increase** in ECN **feedback resolution**.

Refine.dev - YC 23 Remote
Machine Learning Engineer *March 2025 – May 2026*
- Engineered production observability pipeline to detect **attention sinks** degradation in **long-context LLMs**
- Architected a distributed **code indexing system** using Pinecone and Redis caching, achieving **sub-100ms latency retrieval** across **150+ repositories** and reducing **LLM token consumption by 20%**.
- Optimized inference throughput to **2000+ token/second** by implementing **asynchronous API orchestration**, and **load-balancing** across multiple model providers.
- Designed **multi-agent system** with **cost-aware routing layer** and real-time prompt complexity classification.

Index Tax & Financial Remote
Lead Software Engineer *December 2024 – Present*
- Engineered **full-stack platform** for tax financial services firm with **Django REST**, **Nuxt.js**, **Electron.js**, and **Expo**, reducing CPA workflow time by **50%** through automated data aggregation and cross-referencing.
- Architected fully autonomous **RAG pipeline** with **Redis** caching, **Celery** worker distribution, and custom **vector database** to process **500GB+** of client tax documents concurrently.
- Designed **concurrency control system** with **dynamic resource allocation** across **2 GPUs** and **24 cores**.

Projects & Awards

EdgeCart – AI Food Waste Prevention | *Python, OpenCV, ResNet, Gemini API* November 2025
- **4x Winner @ HackPrinceton (Capital One, Knot API, Gemini, YC Runner-Up)**: Developed an edge-computing vision system to detect produce freshness using custom ResNet-50 models
- Implemented high-precision **blemish segmentation** using Google Gemini, improving detection accuracy by **19.5%**
- Implemented **intelligent discount matching** using **Knot API** and **xAI** to target customers, cutting waste by **180M tons annually**

TempoRoll – Real-Time EEG Music Therapy | *Python, PyTorch, Cerebras Systems* September 2025
- **1st Place @ HackMIT (Cerebras Track)**: Engineered a real-time brainwave classification system using a hybrid CNN/ResNet architecture
- Optimized Wafer-Scale Engines inference, achieving **85% accuracy** on EEG streams with **sub 200ms latency**
- Integrated wireless **BLE** EEG communication and generated personalized music to guide brainwaves to baseline

USACO Gold Level (Platinum Division) February 2023

11x Hackathon Winner (Harvard, MIT, Cornell, Princeton 4x, McGill, Brown 2x, Yconic ai) 2025 - Present

Technical Skills

Languages: Python (Advanced), C++, Java, TypeScript, JavaScript, SQL, R, C#
AI & Machine Learning: PyTorch, TensorFlow, vLLM, RAG, Vector Search, Transformers, BERT, CNNs, ResNet
Backend & Infrastructure: FastAPI, Django, Flask, Docker, Kubernetes, Nginx, PostgreSQL, Redis, MongoDB, CI/CD
Web Technologies: Next.js, React, Vue.js, Nuxt.js, Node.js, Nginx, WebSockets, REST APIs
Spoken Languages: English, Chinese (Mandarin and Cantonese)